Syed Tanseer Shah

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Education

S.R.M. Institute of Science and Technology — Chengalpattu, TN
M.Tech (Integrated), CSE – Cognitive Computing; CGPA: 9.24/10 — *2027*

St. Joseph's College — Allahabad, UP
Class XII (ISC), PCM with Computer Science; 93.25% — *2021*

St. Joseph's College — Allahabad, UP
Class X (ICSE), PCMB; 91% — *2019*

Experience

Hindustan Aeronautics Limited (HAL) — June 2026
Intern – Overhaul Division — *Bangalore, Karnataka*

- Developed a YOLO-based vision program to auto-detect aircraft engine parts needing maintenance
- Worked with senior engineers and technicians to turn part-condition criteria into a detection pipeline
- Documented model outputs and part conditions to support quality assurance and airworthiness reporting

Projects

Multimodal AI Psychiatric Pre-Screening Assistant — Oct. 2025 – Present
Python, PyTorch, Transformers, OpenCV, Librosa, Flask, React, Docker

- Built a 3-modality system (NLP, acoustic, vision) for observation-only behavioral analysis in interviews
- Implemented 5 modules: ASR, sentiment analysis, vocal biomarkers, facial emotion, gaze tracking
- Deployed via Flask APIs and Dockerized services with a React clinician dashboard for behavioral timelines

Edge AI Air Quality Prediction & Preventive Purification System — Dec. 2025 – Present
ESP32, TensorFlow Lite Micro (TinyML), Arduino IDE, React, Node.js

- Built an IoT system with 3 sensors (MQ135, BME280, PMS5003) and an on-device TinyML AQI model
- Made predictions run on-device with no cloud dependency, giving low-latency AQI forecasts on an ESP32
- Built a React dashboard for live and historical AQI trends and a relay that auto-triggers purification

Face Recognition System — 2025
Python, Deep Learning, OpenCV, Docker

- Achieved **99.38% accuracy** on the Labeled Faces in the Wild (LFW) benchmark using dlib models
- Built a Dockerized CLI for batch image processing, optimized with Python multiprocessing

CampusKart — 2025
React, Node.js, Express.js, MongoDB, REST APIs, JWT Authentication

- Built a full-stack campus e-commerce platform with JWT auth, REST APIs, and search/filtering

Technical Skills

Languages: Python, Java, C
Domains: Machine Learning, Deep Learning, Computer Vision, NLP, Multimodal AI, Data Science, Embedded/TinyML
Frameworks & Tools: PyTorch, TensorFlow Lite Micro, OpenCV, Transformers, Flask, React, Node.js, Docker, MongoDB, REST APIs
Languages Known: Hindi (Native), English (Native), Japanese (Elementary)